greecon
SH.P.K

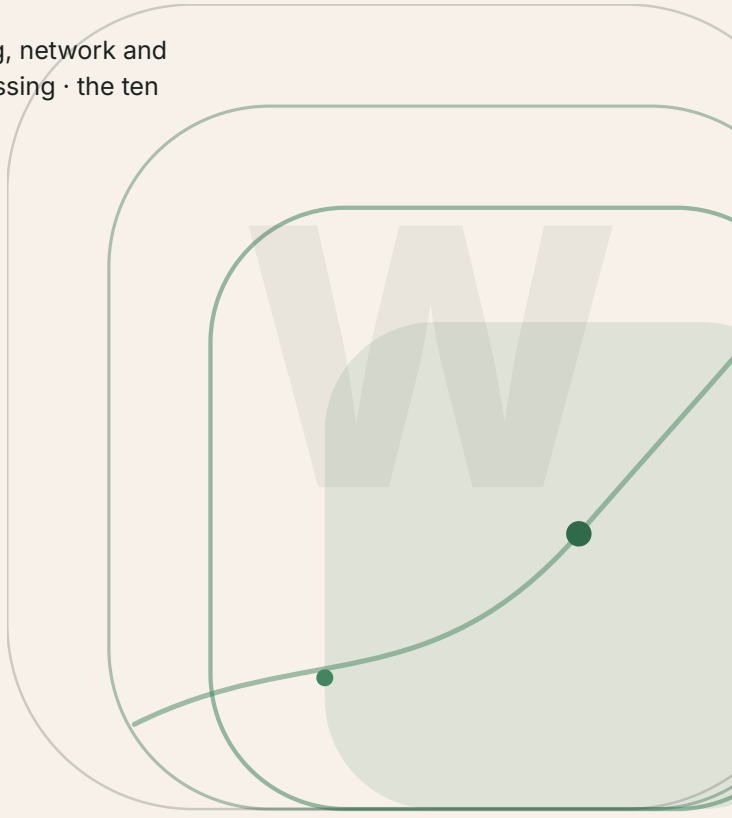
W

SITE PACK W

Water tank and pump station

Tank, pump and the pipe it feeds. Everything for one kind of site in one pack: what to buy, the survey, the diagrams, install day step by step, the platform steps, the commissioning tests and the handover sheet.

FOR	The project manager at this kind of site, the electrician, and the customer's owner at handover
PRINT	The whole pack, for the site in front of you. Tick the boxes as you go
COVERS	Bill of materials · survey · system, wiring, network and enclosure diagrams · install day · addressing · the ten setup steps · commissioning · handover



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SITE PACK W

Water tank and pump station

Tank, pump and the pipe it feeds. A complete pack for one kind of site, in the order of the work: what to buy, the survey, the drawings, the office, install day step by step, the platform's setup list, the commissioning tests and the handover. Print all of it and tick the boxes as you go. The electrician does everything on the mains side; you do the low voltage and the software; Greecon only reads.

At a glance

WHO IT SUITS
A water tank and pump station: a tank, a pump and the pipe it feeds, with the pump's electricity use if wanted. The simplest site, and a good first rehearsal.

WHAT IT MEASURES
The level in the tank, the line pressure, the pump's flow (with a flow meter), whether it runs and, optionally, the pump's electricity use.

PLANNING ESTIMATE
About EUR 650 to 1,300.

TIME
Survey about 1 hour. Install day 4 to 6 hours with the electrician. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Water tank and pump station

STANDARD SETTINGS
Unit numbers: water module 3, optional pump meter 4. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode.

The order of the work: survey, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the 4 to 20 mA signal wires, and the 12 or 24 V supply from a plug-in supply.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

How to read this pack

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, and the pump's run contact.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure W-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Water tank and pump station** makes two devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow	Water I/O module, unit 3: the flow meter's 4 to 20 mA (optional)	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running	Water I/O module, unit 3: the pump starter's spare contact	on or off	None
Energy consumption (optional)	Pump meter (optional): an Eastron SDM230 on the pump's feed, unit 4	kW	None

Without the flow meter, or without the pump's running contact, "the pump is running but no water is flowing" cannot be raised, and the handover should say so.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Analog input module**	Modbus RTU, at least 4 channels, 4 to 20 mA, 7 to 36 V DC	1	30 to 60 (basic) or 150 to 250 (industrial)
☐	**Pump run input**	A Modbus RTU digital input (dry contact) module, or one spare channel of a combined module	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200
☐	**Single-phase meter** *(optional)*	Eastron SDM230, on the pump's feed, to measure its electricity. *To measure the pump's electricity.*	0 or 1	35 to 60

About EUR 650 to 1,300. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in the Hardware Reference (PDF 03).

The project manager brings

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to talk to a meter alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch
- ☐ Spare fuses and terminals
- ☐ A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the electrician if the plan needs one. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

Ask, in this order

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or water? Write it in their words. It is the reason they buy.
3. Where are the water tank, the pump and its starter? How tall is the tank?
4. **What is the pipe's pressure range?** It decides the pressure transmitter's range.
5. **What does the pump do when the tank is low, and when it is full?** And who is told today when something goes wrong?
6. Is there a network socket in the plant room, or cellular signal? (Look at your phone there.)
7. Is there a power socket for the box, and is it on a circuit that stays on?
8. Does the owner want the pump's own electricity measured (an optional single-phase meter on the pump's feed)?
9. Who is the customer's electrician, and can they be there on install day?

Look at, and photograph

- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.
- Where the level and pressure transmitters can be fitted: the pressure transmitter on the pipe through an isolating valve; the level transmitter at the bottom of the tank.
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable.
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT W · WATER TANK AND PUMP STATION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Tank: height (m), water source; drinking water?	
Pump (kW); starter location; what it does when the tank is low and when it is full	
Pipe pressure range (bar), so the pressure transmitter's range	
Sensors needed: level, pressure, flow, pump running (and where)	
Pump meter wanted?	☐ Yes ☐ No
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit W and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers. Mains work is inside the dashed red box and is the electrician's.

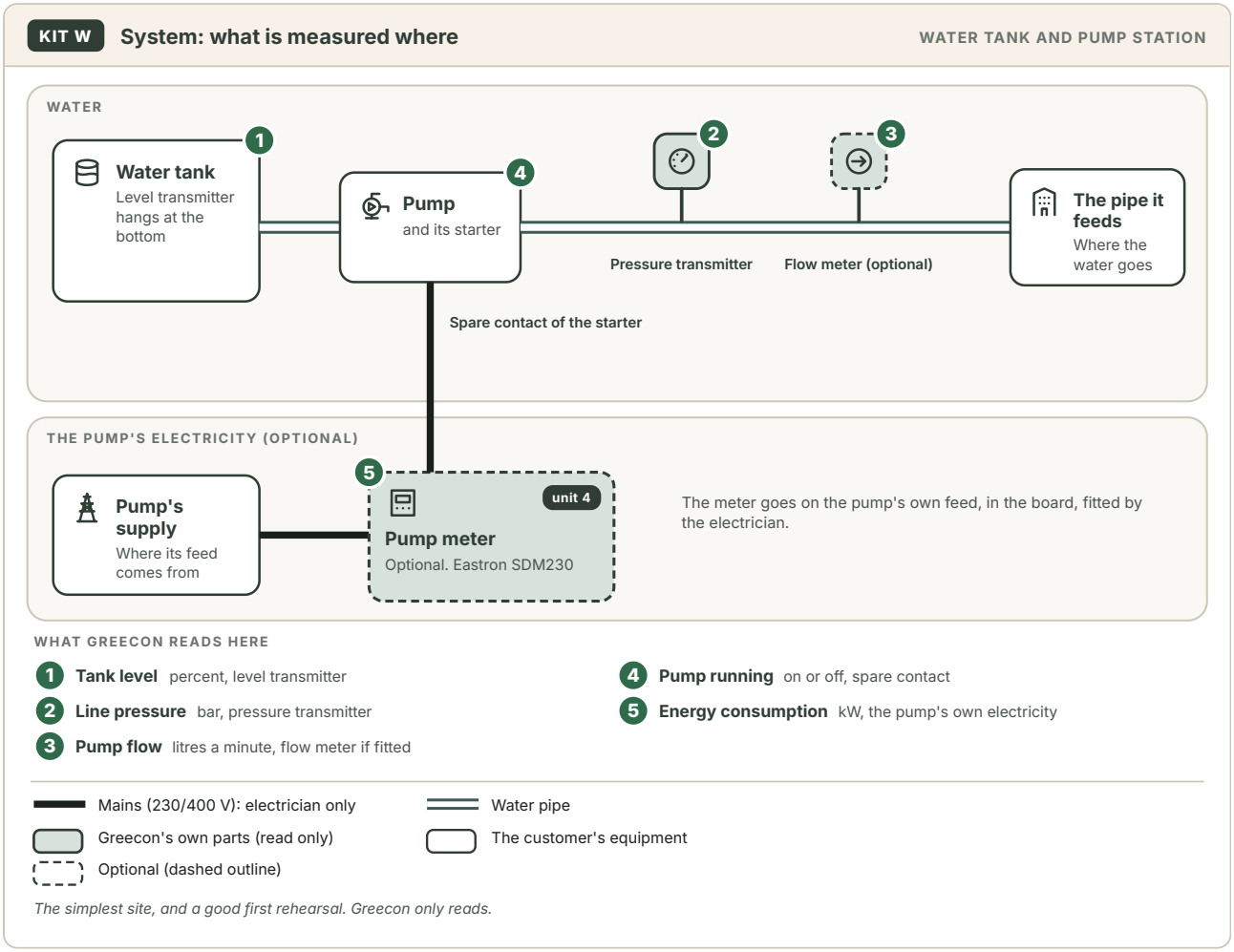
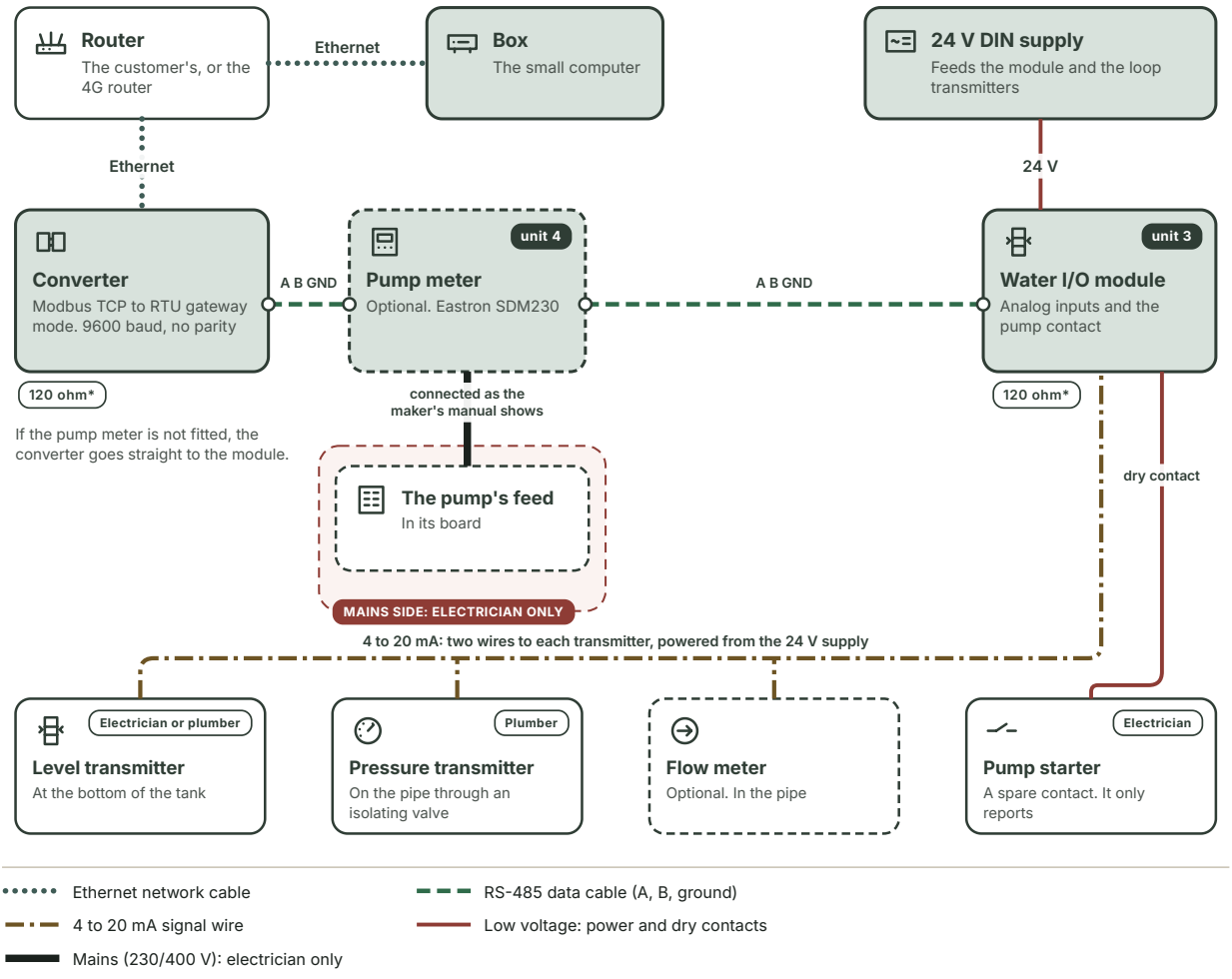


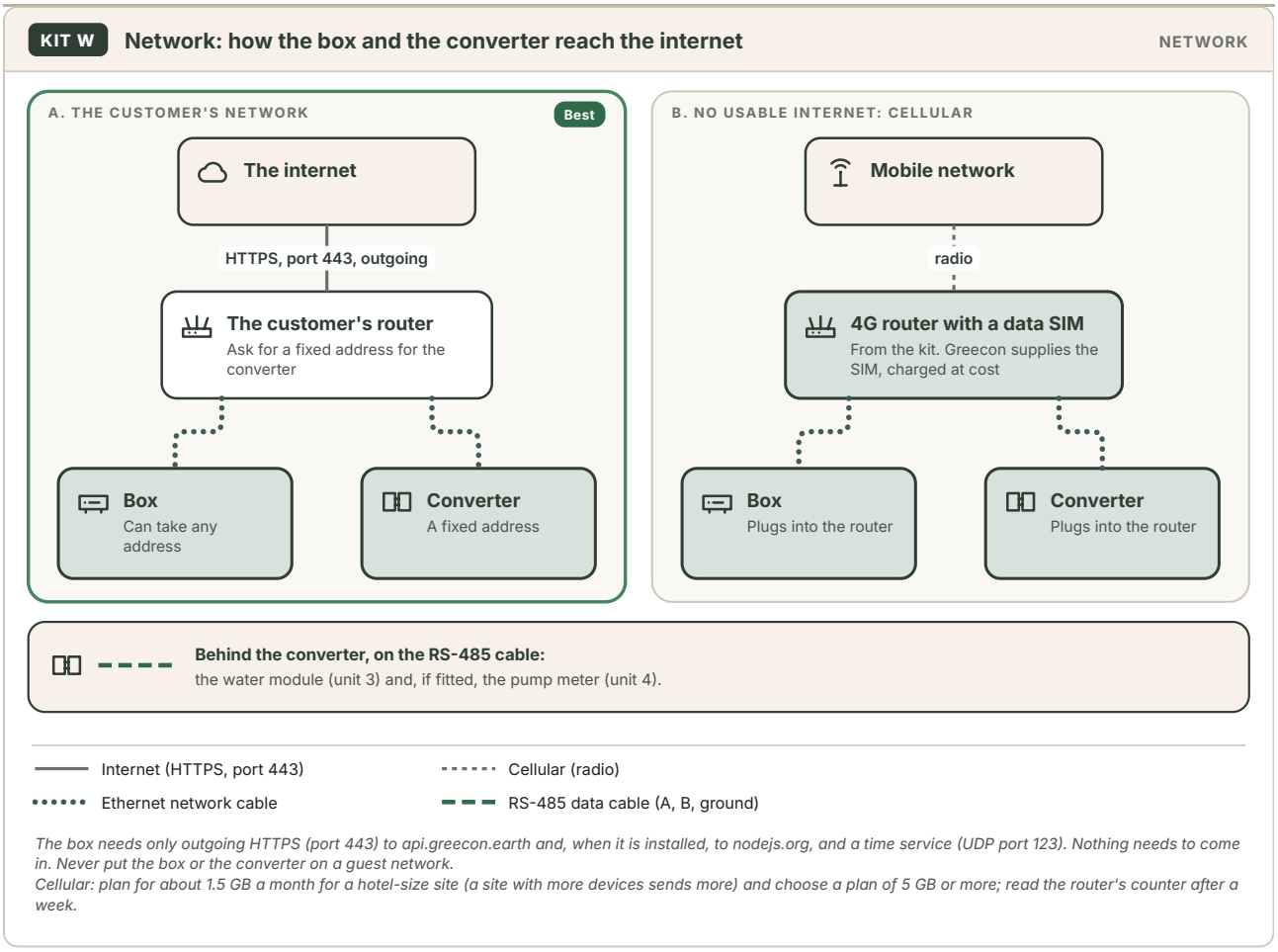
Figure W-1. Kit W, water tank and pump station. System: what is measured where. The dashed parts are optional.

KIT W **Wiring: the data cable and the signal wires** BUS AND SIGNALS



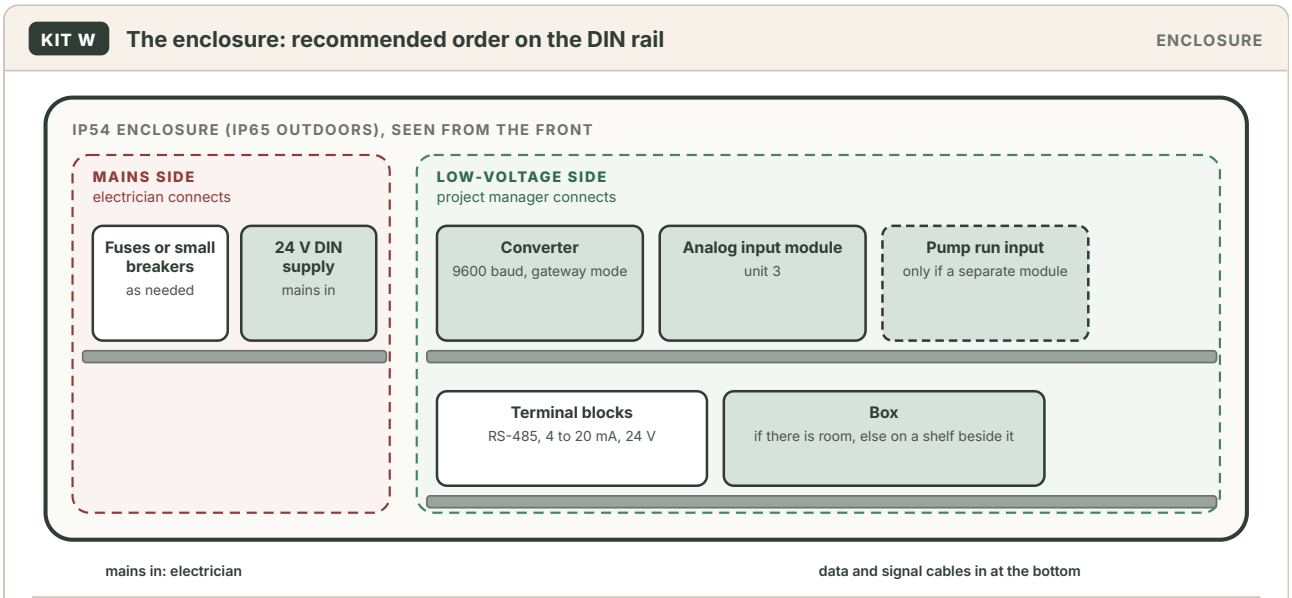
Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The pump meter is connected as its maker's manual shows. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-2. Kit W. Wiring: one RS-485 cable from the converter to the water module (and to the pump meter, if fitted), with the 4 to 20 mA loops and the pump contact into the module.



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure W-3. Kit W. Network: the box and the converter on the customer's router (best) or on a 4G router.



The optional pump meter sits in the board that feeds the pump, not in this enclosure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-4. Kit W. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

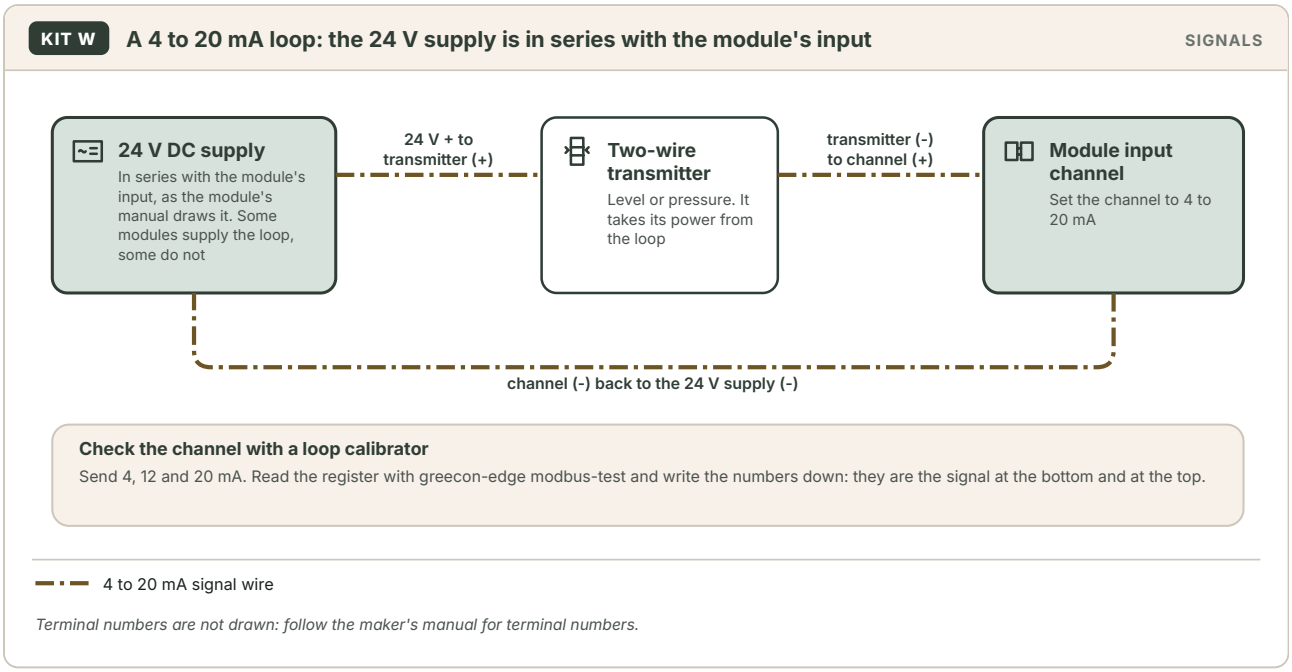


Figure W-5. Kit W. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

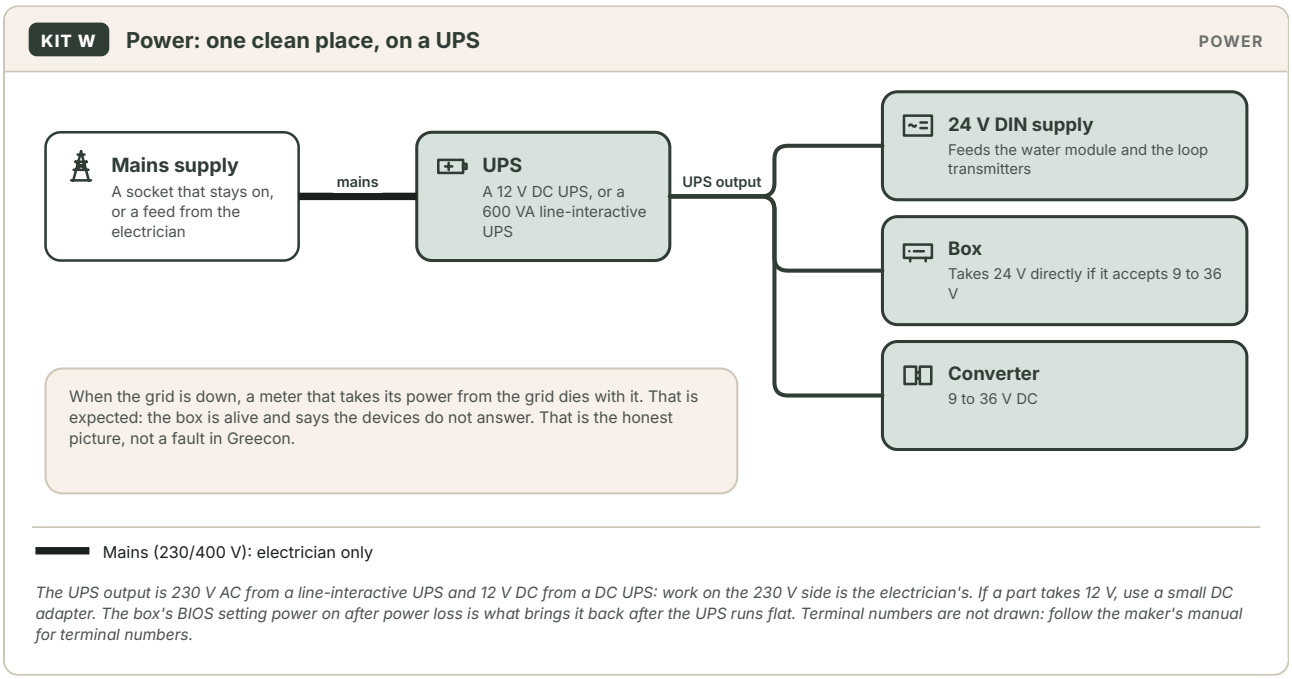


Figure W-6. Kit W. Power: the UPS feeds the box, the converter and the 24 V supply.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box. PROJECT MANAGER

In Greecon

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Water tank and pump station**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.
- [ ] **Get the install line:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`. Press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> **STOP** **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

The box on your desk

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in the Hardware Reference (PDF 03), section 4.3.
- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`

- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit and the printed pack.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the meters: the Project Manager Field Manual (PDF 02), section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN / PROJECT MANAGER	**Walk the site together.** Agree where each sensor goes. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the tank, the pump and the starter. Have the router and a socket that stays on ready, and the person who can give the converter a fixed address.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the board to be worked on.
☐	4	ELECTRICIAN	**Connect the pump's run contact** (a spare auxiliary contact of the starter) to the digital input of the water module. Never to a control circuit: the contact only reports.
☐	5	ELECTRICIAN	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts. The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve.
☐	6	ELECTRICIAN	**Fit the pump meter** (only if wanted) on the pump's feed, as its maker's manual shows.
☐	7	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the water module, the 24 V supply and the UPS (the box too, if there is room, otherwise on a shelf beside it). Figure W-4.
☐	8	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the water module, and on to the pump meter if there is one: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. The electrician connects its end at the pump meter's A and B terminals. Figure W-2.
☐	9	PROJECT MANAGER	**Wire the 4 to 20 mA signals:** two wires from each transmitter to the water module, with the 24 V supply in series with the module's input, as the module's manual draws it. Figure W-5.
☐	10	PROJECT MANAGER	**Set the water module:** unit number 3, and every channel you use to 4 to 20 mA.
☐	11	PROJECT MANAGER	**Set the converter:** Modbus TCP to RTU gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502, and a fixed address on the site's network. Write the address down and photograph the settings page.
☐	12	PROJECT MANAGER	**Connect the network:** plug the box and the converter into the customer's router, or into the sockets of the 4G router if the site has no usable internet. Figure W-3.
☐	13	ELECTRICIAN / PROJECT MANAGER	**Photograph before closing:** the wiring, the converter's settings, each device's label. Then close the board.
☐	14	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.

	#	Who	Do this
☐	15	PROJECT MANAGER	**Set the pump meter at its own display** (if fitted): unit number 4, speed 9600 (the SDM230's speed depends on the version: the V1.6 manual says 2400 as delivered), parity none. Write it down and photograph the display.
☐	16	PROJECT MANAGER	**Power up** the UPS, the 24 V supply, the converter and the box.
☐	17	ELECTRICIAN	**Stay until commissioning is done.**
☐	18	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list.

Addresses and registers

Devices behind one converter share its IP address and differ only by unit number. Write the numbers down here as you set them; you type them into the setup list in step 3.

Device in Greecon	Make and model	Unit number	IP address (the converter's)	Port
Water I/O module		3		502
Pump meter (optional)	Eastron SDM230	4		502

The pump meter, if there is one

The template fills in its register: total active power, input register 12 on the SDM230 (a manual's 30013; counted from zero), a 32-bit floating point number with the high half first, in watts, multiplied by 0.001 to make kW. Test it from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 4 --type input --address 12 --data-type float32
```

The water module

One form for each of its four readings (step 4 of the setup list). The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero. The module's own manual gives what the register holds at 4 and at 20 mA: check those two with a loop calibrator (send 4, 12 and 20 mA, read the register with `greecon-edge modbus-test`) and write what you read.

Reading	Register kind	Address	Conversion	Register at 4 mA	Register at 20 mA	Reading from bottom to top
Tank level			A straight line			0 to 100 (rule below)
Line pressure			A straight line			0 to the range, in bar
Pump flow			A straight line			0 to the range, in litres a minute
Pump running			Multiply and add			on or off

When the transmitter's range is longer than the tank is high: reading at the top = range ÷ tank height × 100. A 0 to 5 m transmitter in a tank 3 m high: 5 ÷ 3 × 100 = 166.7. At the tank's full height the reading is then 100. Delete what the site does not have (no flow meter, no pump meter): open **Equipment and readings**, find it under **Readings** (or its device under **Equipment**) and press **Remove**. A reading with no register holds the setup list at step 4.

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to ten steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself when it sees them work: you press buttons only for what a person must check. A step opens to show its detail and stays as you leave it.

	Step	Title on the setup list	Done by
☐	1	**Prices from the customer's bills**	Project manager (optional)
☐	2	**Install the box**	Project manager
☐	3	**Where each device is on the network**	Project manager
☐	4	**Which register holds which reading**	Project manager
☐	5	**Every device answers the box**	The platform, by itself
☐	6	**Readings arrive and look right**	The platform, by itself
☐	7	**Someone is told when the site needs attention**	Project manager
☐	9	**Alert limits agreed with the customer**	Project manager
☐	10	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

Not on this kit. The platform leaves out a step that does not apply to the site, so the list for kit W has no step 8 (Grid meter direction checked). The steps keep their order; the count on the setup page is out of the steps that do apply, not out of 10.

Step 1 Prices from the customer's bills

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

Step 2 Install the box

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	- *Waiting for the install command to be run on the box.* The line has not been run yet. - *The box has its credential but has not reported yet.* Wait a minute and check the network cable. - *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. - *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

Step 3 Where each device is on the network

PROJECT MANAGER

DO	Under each device, enter its **IP address** (the converter's address), its **Port** (502 unless the converter's page says otherwise) and its **Unit number** (the number you gave the device, 0 to 247). Devices behind one converter share its IP address and differ only by unit number. Press **Save** under each. The box picks up a change within a minute.
YOU SHOULD SEE	**Done**: *Every device has an address,* and each device reads like *192.168.1.50:502, unit 1*.
IF YOU DO NOT	*N devices still need an address.* A wrong address is refused in words, for example *The address must be an IP address such as 192.168.1.50, or a host name.*

Step 4 Which register holds which reading

PROJECT MANAGER

DO	For equipment Greecon knows (the Eastron meters in the template) the registers are already filled in: check them against the meter's manual only if a reading looks wrong. For anything else, press **Choose the register** under the reading's name and fill in: the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero, so 30053 is address 52 and 40001 is address 0), the kind of number (and, for two-register numbers, which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.*
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *No register chosen.* A reading the site does not have (a flow meter that was not fitted, a battery that does not exist) holds the list at this step: open the site, find it under Equipment, in Devices or Reading points, and press Delete.

Step 5 Every device answers the box

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings)*. On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2)*. Look in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address. Nothing answers at all: the IP address or port is wrong, the converter is off, or it is on another network than the box. Some readings work and others do not: a register address is wrong for that device. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

Step 6 Readings arrive and look right

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.*
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *237 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* That is almost always a wrong scale, a wrong register or a wiring fault.

Step 7 Someone is told when the site needs attention

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	▪ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ▪ *Nobody is set to be told:* add the people. ▪ *Nobody on the list has an email address* (or a phone number): add one for each person. ▪ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ▪ *Test messages are limited to 6 an hour.* Wait and try again.

Step 9 Alert limits agreed with the customer

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

Step 10 Commissioned

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 10) with a note. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running. The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail
Pump meter's display against Greecon (if fitted). Read the display and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told and how to reach them, what was commissioned and by whom, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); the site's own page under **Sites**, with the tank level, the pressure and whether the pump runs; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT W · WATER TANK AND PUMP STATION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Alert limits agreed (tank Critical below, Low below; no-flow level)	
Without a flow meter or the pump's run contact the no-flow alert cannot be raised: said to the owner	☐ Said ☐ Not applicable
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, the site's page, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
The pump shows running when it is stopped (and the no-flow alert, an Act now, opens while the pump is stopped)	The pump's running contact is inverted	The electrician swaps the contact's normally-open and normally-closed wires, or uses the opposite contact. Until it is right the no-flow alert cannot be trusted
The tank level is wrong (above 100 percent, or does not move)	A wrong reading at the top for the tank's height, a blocked vented cable, or the transmitter not at the bottom of the tank	Re-enter the reading at the top (range ÷ tank height × 100). Check that the transmitter hangs at the bottom with its vented cable free and fixed. Send 4, 12 and 20 mA with the loop calibrator and compare
The pump meter or the water module does not answer	A wrong unit number, or a speed that does not match the converter. An SDM230 can be delivered at 2400 baud (the V1.6 manual), not the converter's 9600; meters are delivered as unit 1	Read the unit number and speed from the meter's own display and set 4 and 9600 (parity none). Then check A and B, the converter's gateway mode and its IP address

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address.